Complementary Materials for "Sky-Ear: A Unmanned Aerial Vehicle-Enabled Victim Sound Detection and Localization System "

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Appendix A

Proof of Theorem 1: Let TDoA_m be TDoA of the anomaly signal at the microphone array $\mathcal{M} \setminus 0$. Herein, we define TDoA_m as the time delay of received audio from the center microphone to the circular microphones m ($\forall m \in \mathcal{M} \setminus 0$). Using Generalized Cross-Correlation with Phase Transform, TDoA_m ($\forall m \in \mathcal{M} \setminus 0$) is calculated by

$$\mathsf{TDoA}_m = \arg\max_{\tau} \int_{-\infty}^{\infty} \frac{\mathcal{F}(a'_m)\mathcal{F}^*(a'_0)}{|\mathcal{F}(a'_m)\mathcal{F}^*(a'_0)|} e^{j2\pi f\tau} \mathrm{d}f, \quad (1)$$

where $a'_m = a_m[t_{\text{trig}} - \tau_{\text{w}}, t_{\text{trig}}]$, $a'_0 = a_0[t_{\text{trig}} - \tau_{\text{w}}, t_{\text{trig}}]$, and $\mathcal{F}\{\cdot\}$ is the Fourier transform and $\mathcal{F}^*\{\cdot\}$ is its complex conjugate. The geometric relationship among $\mathbf{r}_0$, $\mathbf{r}_m$, $\overrightarrow{\mathsf{DoA}}$, and TDoA_m is as follows:

$$\forall m \in \mathcal{M} \setminus 0 : \mathsf{TDoA}_m \cdot v_{\text{s}} = (\mathbf{r}_m - \mathbf{r}_0)^T \cdot \overrightarrow{\mathsf{DoA}}, \quad (2)$$

where v_{s} is the speed of sound. Eq. (2) includes $M-1$ equations for $M-1$ audio sequences recorded from $\mathcal{M} \setminus 0$. To derive $\overrightarrow{\mathsf{DoA}}$ from Eq. (2), we can minimize the squared projection error of $M-1$ equations, that is,

$$\min \|\mathbf{G} \cdot \overrightarrow{\mathsf{DoA}} - \mathbf{V}\|^2, \quad (3)$$

where $\mathbf{G} = \begin{bmatrix} (\mathbf{r}_1 - \mathbf{r}_0)^T \\ (\mathbf{r}_2 - \mathbf{r}_0)^T \\ \vdots \\ (\mathbf{r}_{M-1} - \mathbf{r}_0)^T \end{bmatrix}$ and $\mathbf{V} = [\mathsf{TDoA}_1 \cdot v_{\text{s}}, ..., \mathsf{TDoA}_{M-1} \cdot v_{\text{s}}]$. Solve Eq. (3) using the Least Squares method, the analytical solution of $\overrightarrow{\mathsf{DoA}}^*$ is given by

$$\overrightarrow{\mathsf{DoA}}^* = (\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T\mathbf{V}. \quad (4)$$

Finally, $\overrightarrow{\mathsf{DoA}}^*$ is normalized to unit length. Substituting $\mathbf{G}$, $\mathbf{V}$, and TDoA_m into Eq. (4), we have Theorem 1. ■

Appendix B

Proof of Theorem 2: Due to inevitable measurement error, e.g., the dynamic acoustic environment and varying sound propagation paths, these observed directions form non-intersecting skew lines, leading to different confidence in localizing the victim. To this end, we introduce a confidence weight w_k for each observation point k. Then, an optimal point $\mathbf{s}^*$ is obtained by minimizing the sum of the weighted squared orthogonal distances to all K observed directions, i.e.,

$$\mathbf{s}^* = \arg\min_{\mathbf{s}} \sum_{k=1}^{K} w_k \left\| \overrightarrow{\mathsf{Proj}}_k (\mathbf{s} - \mathbf{p}_k) \right\|^2, \quad (5)$$

where $\overrightarrow{\mathsf{Proj}}_k = \mathbf{I} - \overrightarrow{\mathsf{DoA}}_k^* (\overrightarrow{\mathsf{DoA}}_k^*)^T$ represents the orthogonal projection matrix that projects any vector onto the orthogonal complement of $\overrightarrow{\mathsf{DoA}}_k^*$ with $\mathbf{I}$ being the 3×3 identity matrix. The confidence weight w_k is defined by the normalized peak magnitude of the time-domain cross-correlation function calculated in Eq. (1), which is given by

$$w_k = \frac{1}{M-1} \sum_{m \in \mathcal{M} \setminus 0} \max_{\tau} \left| \mathcal{F}^{-1} \left\{ \frac{\mathcal{F}(a'_m)\mathcal{F}^*(a'_0)}{|\mathcal{F}(a'_m)\mathcal{F}^*(a'_0)|} \right\} \right|, \quad (6)$$

where $\mathcal{F}^{-1}\{\cdot\}$ denotes the inverse Fourier transform. Solve the problem in Eq. (5), Theorem 2. ■

Appendix C

The workflow of Encoder: For $i \in \mathcal{V}$, each 2D patch $X_i^{P \times P}$ is first flattened into a 1D raw vector $\mathbf{x}_i^{1 \times P^2}$. To adapt to a uniform dimension D in the encoder, we multiply the flattened vector by a weight matrix $\mathbf{E} \in \mathbb{R}^{P^2 \times D}$ to convert the P^2 raw pixels into a D-dimensional feature vector. Next, to retain the 2D spatial geometry of the original Mel-spectrogram, a positional encoding $\mathbf{p}_i^{\text{enc}}$ is superimposed onto each feature vector, denoted as $\mathbf{x}_i\mathbf{E} + \mathbf{p}_i^{\text{enc}}$. To combine each feature vector, we have added a classification label $\mathbf{z}_{\text{cls}} \in \mathbb{R}^{1 \times D}$ before the sequence and concatenated these individual feature vectors to form a one-dimensional input sequence. The fully assembled input sequence $\mathbf{Z}_{\text{in}}$ is formulated as

$$\mathbf{Z}_{\text{in}} = [\mathbf{z}_{\text{cls}}; \mathbf{x}_{v_1}\mathbf{E} + \mathbf{p}_{v_1}^{\text{enc}}; \mathbf{x}_{v_2}\mathbf{E} + \mathbf{p}_{v_2}^{\text{enc}}; \ldots; \mathbf{x}_{v_{|\mathcal{V}|}}\mathbf{E} + \mathbf{p}_{v_{|\mathcal{V}|}}^{\text{enc}}], \quad (7)$$

where $v_1, v_2, ..., v_{|\mathcal{V}|}$ go through all index from $\mathcal{V}$. Finally, $\mathbf{Z}_{\text{in}}$ is input into the Transformer block $f_{\text{enc}}(\cdot)$, which calculates the correlation between each pair of feature vectors by integrating the complete information in the sequence and representing comprehensive acoustic environmental information. Thus, we have the final feature sequence in Eq. (2). ■

Appendix D

The workflow of Decoder: We constuct an intermediate feature sequence $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_N]$. For $i \in \{1, 2, ..., N\}$, $\mathbf{h}_i$ is padded by

$$\mathbf{h}_i = \begin{cases} \mathbf{z}_i\mathbf{W}_{\text{dec}}, & \text{if } i \in \mathcal{V}. \\ \mathbf{t}_{\text{mask}}, & \text{if } i \in \mathcal{U}. \end{cases} \quad (8)$$

When $i \in \mathcal{V}$, $\mathbf{z}_i$ is extracted from $\mathbf{Z}_{\text{enc}}$ by multiplying a feature dimensional weight matrix $\mathbf{W}_{\text{dec}}$; when $i \in \mathcal{U}$, $\mathbf{t}_{\text{mask}}$ serves as a shared learnable placeholder, indicating that the patch has been masked and requires subsequent reasoning. To provide

the original spatial coordinates of both the visible features and mask patches, a positional encoding $\mathbf{P}_{\text{dec}}$ is superimposed onto the sequence $\mathbf{H}$. Furthermore, the updated classification label $\mathbf{z}'_{\text{cls}}$ is prepended to the sequence's beginning. This updated label is obtained by projecting the encoder's output classification label through the same dimension-matching weight matrix, yielding $\mathbf{z}'_{\text{cls}} = \mathbf{z}_{\text{cls}}\mathbf{W}_{\text{dec}}$. This sequence $\mathbf{Z}_{\text{dec}}$ assembled by the **Decoder**, is formulated as

$$\mathbf{Z}_{\text{dec}} = [\mathbf{z}'_{\text{cls}}; \mathbf{h}_1; \mathbf{h}_2; \ldots; \mathbf{h}_N] + \mathbf{P}_{\text{dec}}. \tag{9}$$

Loading the **Decoder** sequence $\mathbf{Z}_{\text{dec}}$ into the Transformer block $f_{\text{dec}}(\cdot)$, the correlation sequence $\mathbf{Z}_{\text{out}} \in \mathbb{R}^{1\times(N+1)}$ between each vector can be calculated, which is given by

$$\mathbf{Z}_{\text{out}} = f_{\text{dec}}(\mathbf{Z}_{\text{dec}}) = [\tilde{\mathbf{z}}_{\text{cls}}; \tilde{\mathbf{z}}_1; \tilde{\mathbf{z}}_2; \ldots; \tilde{\mathbf{z}}_N]. \tag{10}$$

where $\tilde{\mathbf{z}}_n$ ($n \in \{1, 2, ..., N\}$) represents the n-th estimated feature vector with the missing acoustic fragments being successfully recovered. Next, we need to fill the **decoder** with the sequence $\mathbf{Z}_{\text{out}}$ back to a 2D Mel-spectrogram. To achieve this, the classification label $\tilde{\mathbf{z}}_{\text{cls}}$ is discarded, each D-dimensional feature vector $\tilde{\mathbf{z}}_n$ from the sequence $\mathbf{Z}_{\text{out}}$ is first converted into P^2 numerical values. Implemented through a weight matrix $\mathbf{W}_{\text{pred}}$ and an offset vector $\mathbf{b}_{\text{pred}}$. Specifically, $\mathbf{W}_{\text{pred}} \in \mathbb{R}^{D\times P^2}$ is responsible for mapping the D-dimensional hidden information back into P^2 distinct numerical intensities. And $\mathbf{b}_{\text{pred}} \in \mathbb{R}^{P^2}$ is applied to shift the predicted values to their correct physical baseline. The resulting 1D array of P^2 predicted values $\hat{\mathbf{x}}_n$ for the n-th patch is calculated as

$$\hat{\mathbf{x}}_n = \tilde{\mathbf{z}}_n\mathbf{W}_{\text{pred}} + \mathbf{b}_{\text{pred}}, \quad \forall n \in \{1, 2, \ldots, N\}. \tag{11}$$

Finally, to recover the 2D spatial arrangement defined in the initial masking phase, each 1D array $\hat{\mathbf{x}}_n$ is geometrically reshaped back into a $P \times P$ image patch. These N patches are then reassembled according to their original global position indices $\{1, 2, \ldots, N\}$ to form the complete Mel-spectrogram. The final reconstructed image patch $\tilde{X}_n$ is given by

$$\tilde{X}_n = \mathcal{R}(\hat{\mathbf{x}}_n), \quad \forall n \in \{1, 2, \ldots, N\}, \tag{12}$$

where $\mathcal{R}(\cdot)$ denotes the reshaping operator. Substituting Eqs. (8) to (11) into Eq. (12), we have Eq. (3) in our paper. ■